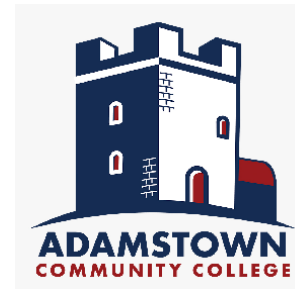




Lesson 03

Inputs and Strings



Part 01

Inputs



Taking user Input

- Up to now we have been using hardcoded Variables
- To make a program dynamic we must take values from the user
- Python has an input Stream and an output stream
- We have used the output stream (`print("Hello Worls")`)
- Now we will use the input stream (`input()`)



Taking user input

- We want to collect the users name and pass it to a variable
- `name = input()`
- `print("Hello",name)`
- The input function takes input from the user



- There is a small problem with `name = input()`
- Code

```
name = input()  
print("Hello", name)
```

Output

```
█
```

- Generally you should prompt the user

```
name = input("Please enter your name: ")  
print("Hello", name)
```

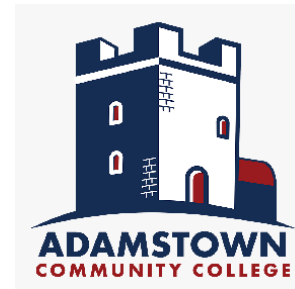
```
Please enter your name: █
```

```
Please enter your name: Alan  
Hello Alan
```



Taking User input

- We have taken in a string
- By default input takes in anything typed as a string
- What if we want a number
- We need to convert the input to an integer using the integer function
- We wrap the input in this function
- `Age = input("Please enter your age")` – Becomes
- `Age = int(input("Please enter your age"))` – Note the brackets
- For a decimal number we use `float()` instead of `int()`



Part 2

Strings



Strings

- Strings in python are immutable
- This means they cannot change
- However there are many things can be done to string

We can add strings together:

```
string1 = "hello"  
string2 = "world"  
string3 = string1 + " " + string2  
print(string3)
```

Output:

hello world

- We can multiply strings

```
string1 = "hello"  
string2 = string1 * 3  
print(string2)
```

- Output:

hellohellohello



- There are functions you can use on strings

- Make the string upper case

```
my_string = "hello"  
my_string = my_string.upper()  
print(my_string)
```

- Output:

HELLO

- Make the string lower case

```
my_string = "HELLO"  
my_string = my_string.lower()  
print(my_string)
```

- Output:

hello

- Count the length of a string

```
my_string = "elephant"  
length_of_string = len(my_string)  
print(length_of_string)
```

- Output:

8

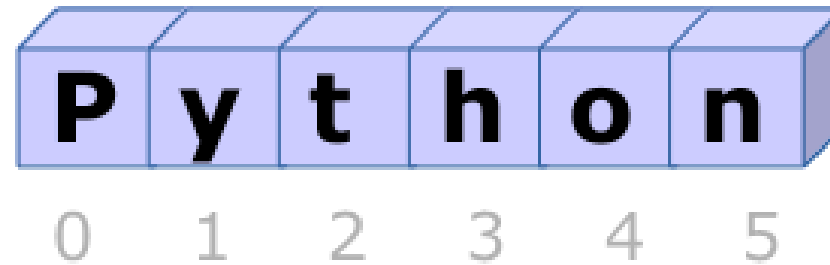


Printing formatted strings

- Concatenation (adding strings together)
- `name = "Alan"`
`age = 31`
`output = "Hello" + name + "you are" + str(age) + "years old."`
`print(output)`
- Why do we wrap the age variable in `str()`
- Age is an integer and needs to be converted to a string

Indexing

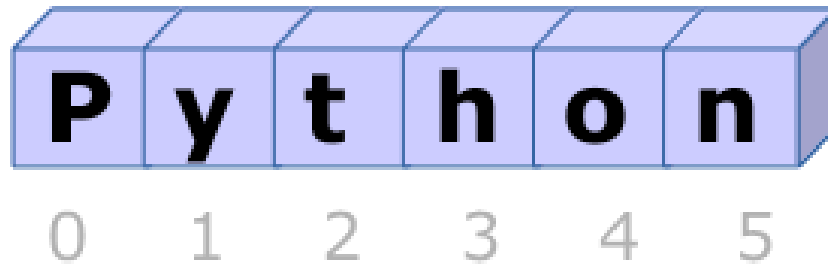
- Strings have an index where each character is assigned a number starting from 0



- String indexes always start at 0
- If we want to print the character at index 3
`myString = "Python"`
`print(myString[3])`
- What letter will be printed

Slicing

- We can use slicing to cut a string into smaller parts



- `myString="Python"`
`print(mystring[0:2])`
- Output: Py
- Why do you think only the first 2 characters printed?
- Try other slicing options to figure it out



- We wanted to print out the characters from index 0 to 2 whu did it only print out the first 2
- When you slice a string you print out up to but not including the last index
- If you want the first 3 then you should `print(myString[0:3])`

Summary



- Today you learned
 - How to take input from a user
 - Adding strings together using Concatenation
 - You can multiply strings using *
 - That strings have indexes that refer to each character
 - That you can extract parts of a string by referring to these index numbers